University of Florida
College of Public Health & Health Professions Syllabus

PHC 6446: Systems Thinking in One Health (3 credit hours)

Spring: 2026
Delivery Format: Online & On-campus
Course Website: PHC6446 Systems Thinking in One Health

Instructor Name: Benjamin Anderson, MPH, PhD, CPH
Meeting Time: Mondays, 9:35am-11:30am
Room Number: HPNP G-112
Phone Number: 352-273-9572
Email Address: ander88@ufl.edu
Office Hours: Mondays, 12pm-1pm or by appointment
Preferred Course Communications: Canvas

Prerequisites
None

PURPOSE AND OUTCOME

Course Overview
In this course, students will engage in a dynamic learning environment to understand how systems thinking can be used to uncover the root causes of complex health problems at the human, animal, and environmental interface, and to identify leverage points where small interventions can lead to significant changes.

Relation to Program Outcomes
Systems Thinking is deemed a core competency of the One Health approach, which employs interdisciplinary problem-solving at the human, animal, and environmental interface. This course is specifically part of the **Core Concentration** requirements of the MHS in One Health and the PhD in Public Health (One Health Concentration) degree programs. Successful completion of this course fulfills the Systems Thinking learning objective.

Course Objectives and/or Goals
The goal of this course is to learn and apply Systems Thinking strategies to mitigate complex health problems at the human, animal, and environmental interface (One Health).

At the end of this course, you will be able to:

1. Introduce the concept of Systems Thinking as it relates to the One Health Approach.
2. Identify the underlying structures, behaviors, and dynamics of different types of complex systems.
3. Explore the relationship between structural and functional elements of a system.
4. Apply mapping techniques (Causal Loop Diagram, Social Network Analysis, etc.) to visualize interconnections and feedback loops within and across various systems.
5. Design mitigation and evaluation strategies targeted at leverage points within a system to maximize effective change.
6. Critically analyze the scientific literature on Systems Thinking.
7. Communicate Systems Thinking concepts effectively with peers.

Instructional Methods

This course will be delivered online with one weekly live synchronous session. The activities carried out in the synchronous sessions will supplement the online learning instruction that will be delivered through the Canvas course site. Synchronous sessions are meant to be highly interactive and provide a space for students to critically discuss the course concepts with their peers. It is expected that all students will attend the synchronous discussion sessions.

Blended Learning

What is blended learning and why is it important?

A Blended Learning class uses a mixture of technology and face-to-face instruction to help you maximize your learning. Knowledge content that, as the instructor, I would have traditionally presented during a live class lecture is instead provided online before the live class takes place. This lets me focus my face-to-face teaching on course activities designed to help you strengthen higher order thinking skills such as critical thinking, problem solving, and collaboration. Competency in these skills is critical for today's health professional.

What is expected of you?

You are expected to actively engage in the course throughout the semester. You must come to class prepared by completing all out-of-class assignments. This preparation gives you the knowledge or practice needed to engage in higher levels of learning during the live class sessions. If you are not prepared for the face-to-face sessions, you may struggle to keep pace with the activities occurring in the live sessions, and it is unlikely that you will reach the higher learning goals of the course. Similarly, you are expected to actively participate in the live class. Your participation fosters a rich course experience for you and your peers that facilitates overall mastery of the course objectives.

DESCRIPTION OF COURSE CONTENT

Topical Outline/Course Schedule

[This list must correspond with the number of weeks that the course is being offered. You must list week or dates and topics (not just a list of topics). You should also list reading assignments if known. Faculty may need to adapt this section according to the level of detail they wish to provide.]

Week	Date(s)	Topic(s)	Assignments/Readings
Module 1 – Introduction to Systems Thinking in One Health			
1	1/12-1/16	Introduction to Systems Thinking in One Health	See Canvas
2	1/19-1/23	Systems Thinking Methodology and Systems Mapping (Miro); Introduce Systems Thinking Project (Presentation and Paper)	**Module 1 Discussion Due** **Application Exercise 1 Due**
Module 2 – Health and Healthcare Systems			
3	1/26-1/30	Defining Health; Global Health Systems	See Canvas
4	2/02-2/06	Global Burdan of Disease (GBD); Measuring Health in a System; SDG Dashboard	**Project Topic Form Due** **Module 2 Discussion Due** **Application Exercise 2 Due** **Quiz 1 Due**
Module 3 – Food Systems			
5	2/09-2/13	Globalization of Food Supplies; Food Insecurity and Health	See Canvas
6	2/16-2/20	Future of Food Production	**Project Outline Due** **Module 3 Discussion Due** **Application Exercise 3 Due**
Module 4 – Biodiversity and Emerging Disease Hotspots			

Week	Date(s)	Topic(s)	Assignments/Readings
7	2/23-2/27	Impact of Climate Change on Biodiversity; Emerging Infectious Diseases	See Canvas
8	3/02-3/06	Future Models of Pandemic Preparedness	**Module 4 Discussion Due** **Application Exercise 4 Due** **Quiz 2 Due**
Module 5 – Socioecological System			
9	3/09-3/13	Introduction to the socioecological system perspective	See Canvas **Module 5 Discussion Due** **Application Exercise 5 Due**
Spring Break – 03/14-03/21			
Module 6 – Applying Systems Thinking to Health Interventions			
10	3/23-3/27	Health Intervention and Prevention Strategies; Monitoring and Evaluation	See Canvas
11	3/30-4/03	Case Study	**Module 6 Discussion Due** **Application Exercise 6 Due** **Quiz 3 Due**
Project Presentations			
12	4/06-4/10	How to give an effective presentation; Workshop Draft Projects	See Canvas
13	4/13-4/17	**Project Presentations** **PHHP Days 2026 – 04/15 and 04/16**	**Quiz 4 Due** **Optional: PHHP Days 2026 Assignment Due**
14	4/20-4/22	**Project Presentations Cont.**	**Project Paper Due**

Course Materials and Technology

There is no required textbook for this course. All assigned readings will be made available through the Canvas course site.

"Instructional materials for this course consist of only those materials specifically reviewed, selected, and assigned by the instructor(s). The instructor(s) is only responsible for these instructional materials."

For the systems mapping requirements, you will need to create a **free** account at www.miro.com.

ACADEMIC REQUIREMENTS AND GRADING

Assignments

Application Exercises – For each module, you will complete an individual application exercise where you will practice applying a knowledge concept or skill covered in class or in the reading. Each application exercise will require a written summary of the applied method. These will be graded using a standard rubric with formative feedback to male improvements throughout the course.

Discussion Boards – For each module, one question will be posted on the Discussion section of the Canvas course site that you will be required to respond to. In addition, you will be required to thoughtfully respond to at least one post

from one of your classmates. The purpose of this activity is to extend the discussion of module topics beyond the classroom and provide opportunities to practice communicating your ideas in written format.

Quizzes – There will be no midterm or final exam in this class. Instead, there will be quizzes covering module material throughout the course (4 in total) via Canvas. Questions may be multiple choice, true/false, and/or short answer, and will cover topics from both assigned videos/readings. Quizzes are to be completed independently, but you may refer to your notes and assigned readings. **AI is not permitted in completing your quizzes responses.**

Systems Thinking Project (Presentation and Paper) – During the course, you will identify and research a complex global health issue focused in one country of your choice. You will then propose an intervention that builds upon the systems thinking concepts and activities conducted throughout the course. This project will culminate in a final presentation structured as an address to an emergency session of the world health assembly and the submission of a research paper.

Grading

Requirement	Due date	% of final grade
Module 1 Discussion	1/26	1%
Application Exercise 1	1/26	5%
Project Topic Form	2/09	2%
Module 2 Discussion	2/09	1%
Application Exercise 2	2/09	5%
Quiz 1	2/09	10%
Project Outline	2/23	2%
Module 3 Discussion	2/23	1%
Application Exercise 3	2/23	5%
Module 4 Discussion	3/09	1%
Application Exercise 4	3/09	5%
Quiz 2	3/09	10%
Module 5 Discussion	3/23	1%
Application Exercise 5	3/23	5%
Module 6 Discussion	4/06	1%
Application Exercise 6	4/06	5%
Quiz 3	4/06	10%
Quiz 4	4/20	10%
Optional: PHHP Days 2026 Assignment	4/20	Replace Lowest AE
Project Presentation Due	4/22	10%
Project Paper Due	4/22	10%

Point system used:

Percentage Earned	Letter Grade
93-100	A
90-92	A-
87-89	B+
83-86	B
80-82	B-
77-79	C+
73-76	C
70-72	C-
67-69	D+
63-66	D
60-62	D-
Below 60	E

Please be aware that a C- is not an acceptable grade for graduate students. The GPA for graduate students must be 3.0 based on 5000 level courses and above to graduate. A grade of C counts toward a graduate degree only if based on credits in courses numbered 5000 or higher that have been earned with a B+ or higher.

Letter Grade	Grade Points
A	4.0
A-	3.67
B+	3.33
B	3.0
B-	2.67
C+	2.33
C	2.0
C-	1.67
D+	1.33
D	1.0
D-	0.67
E	0.0
WF	0.0
I	0.0
NG	0.0
S-U	0.0

More information on UF grading policy may be found at:
https://gradcatalog.ufl.edu/graduate/regulations/#Grades *[Remove if course is an undergraduate level course.]*
https://catalog.ufl.edu/UGRD/academic-regulations/attendance-policies/ [Remove if course is a graduate level course.]

Exam Policy
There will not be exams for this course. Instead, there will be 6 quizzes (one per module) administered with a combination of multiple choice, true/false, and free response questions. These will allow students to see learning progress and receive formative feedback for improvement.

Policy Related to Make Up Exams or Other Late Work

Please note all faculty are bound by the UF policy for excused absences. For information regarding the UF Attendance Policy see the Registrar website for additional details. I encourage you to please reach out to the instructor if circumstances arise where you need to be absent or submit late work. I will try to balance professionalism with compassion so that we can make alternative plans, when necessary, to help you be successful in this course while prioritizing your emotional and physical well-being. If possible, please reach out before missing a class or a quiz or test.

Any requests for make-ups due to technical issues should be accompanied by the ticket number received from eLearning when the problem was reported to them. The ticket number will document the time and date of the problem. Please e-mail your instructor within 24 hours of the technical difficulty if you wish to request a make-up for a Canvas quiz.

It is my intent that all quizzes and tests will be turned in on time (see dates on course schedule). Assessments cannot be made-up except in the case of circumstances that meet the criteria of the University of Florida policy for an excused absence. However, as described above, please reach out if special circumstances occur, and I will work with you as best as possible.

In the event of an absence, inform the instructor as early as possible prior to the class. For all unplanned absences because of accidents or emergency situations, students should contact their instructor as soon as conditions permit. If an absence for one of the above reasons results in missing an exam or quiz, contact the instructor to arrange a make-up exam or quiz, which will typically take place within one week of the due date.

The general policy for late work is that your final points will be reduced by 10% a day (starting immediately after the missed deadline). This penalty can be waived with instructor discretion.

Course Policy Related to Required Class Attendance
To get the most out of the course, it is expected that students will attend weekly sessions. If you need to miss a session, please let me know in advance.

Policy Related to AI Use in This Course
When authorized by the course director, students may use AI technologies in the completion of coursework as long as they cite all such uses by naming the technology and how it was employed. Students assume full responsibility for all content, including errors and omissions. Assistive technology authorized as part of an accommodation for a disability is always permitted.

Course instructors may adjust limitations on AI assistive technology use and must communicate any limitations to students sufficiently in advance of the assignment due date. Failure to cite the use of AI assistive technology, or use of the technology disregarding specific course limitations is considered academic misconduct. **The use of AI on assignments, essays/reflection papers, exams, and quizzes when prohibited by course or college instructions is considered cheating** and students are violating the UF Regulations 4.040 Student Honor Code and Student Conduct Code.

It is important to note that many generative AI models (e.g. ChatGPT, ChatSonic, Google Bard etc) place any information that they are provided with into the public domain. When using such tools, you must therefore ensure that they are **never provided with confidential information**. UF AI systems (e.g., Co-Pilot, NaviGator) should never be provided with confidential information. For the avoidance of doubt, the use of such tools is prohibited for generating

any confidential communications, including, but not limited to, communications relating to patient records, clients, students and intellectual property. You are also reminded that you should always review the terms and conditions of any third-party software you use (e.g. proof-reading tools) to ensure that any data they are provided with is appropriately protected. Always verify information and sources generated by AI tools. AI has been known to generate false information and to cite non-existent sources. Also, because AI-generated text mines people's intellectual property without appropriate credit, this raises ethical concerns.

It is not acceptable to use generative AI for reflective writing, as by its very nature, the process of reflective writing demands that you actively engage in the writing process. Delegating this to a natural language processing algorithm may produce convincing outputs but does not demonstrate development in your professional practice.

Students are responsible for understanding their dynamic data stewardship responsibilities to minimize personal, college, and university risk.

UF Integrated Risk Management – CHATGPT Privacy, Factual Accuracy and Usage Guidelines

ACADEMIC POLICIES & RESOURCES

University academic policies and resources can be found at: https://syllabus.ufl.edu/syllabus-policy/uf-syllabus-policy-links/

STUDENT EXPECTATIONS, ROLES, AND OPPORTUNITIES FOR INPUT

Expectations Regarding Course Behavior

Students are expected to maintain a civil tone and respect the opinions of other people in class and online discussions.

Communications Guidelines

For all course communications to the instructor, please use the "Inbox" feature in Canvas. Weekday daytime (US Eastern Time) emails have the best chances of being answered quickly.

For additional suggestions, please refer to the Netiquette Guidelines:
https://teach.ufl.edu/wp-content/uploads/2020/04/NetiquetteGuideforOnlineCourses.docx

PHHP Student Resources

PHHP's UPTurn (Unified Pathways to Support Wellness) program is a *no-cost mental health and wellness program* that is offered year-round to all PHHP students (undergraduate, graduate and professional level) and students (from any college) who are enrolled in PHHP courses. UPTurn advisors support students on their wellness journeys by curating individualized plans (resources and support) to help them manage academic, social, emotional, and health-related stress.

Interested students are paired with an UPTurn advisor, who meets with each student *virtually* (Zoom, Teams, phone) or *in person* (private office/room in HPNP) for a 45-minute consultation, followed by (if desired):

1. Up to 4 follow-up skills-building visits
2. When needed and appropriate, up to 10 psychotherapy sessions after completion of the 4 follow-up skills-building visits

Note: UPTurn is NOT a crisis/emergency resource. Students who are in crisis are strongly encouraged to use UF's existing crisis support resources, which are listed here: https://counseling.ufl.edu/services/crisis/

Students can learn more about UPTurn and request an appointment here: https://phhp.ufl.edu/student-resources/upturn-wellness-program/

Any questions regarding UPTurn can be directed to upturn@phhp.ufl.edu or (352) 273-6850.

Inclusive Learning Environment

Public health and health professions are based on the belief in human dignity and on respect for the individual. As we share our personal beliefs inside or outside of the classroom, it is always with the understanding that we value and respect diversity of background, experience, and opinion, where every individual feels valued. We believe in, and promote, openness and tolerance of differences in ethnicity and culture, and we respect differing personal, spiritual, religious and political values. We further believe that celebrating such diversity enriches the quality of the educational experiences we provide our students and enhances our own personal and professional relationships. We embrace The University of Florida's Non-Discrimination Policy, which reads, "The University shall actively promote equal opportunity policies and practices conforming to laws against discrimination. The University is committed to non-discrimination with respect to race, creed, color, religion, age, disability, sex, sexual orientation, gender identity and expression, marital status, national origin, political opinions or affiliations, genetic information and veteran status as protected under the Vietnam Era Veterans' Readjustment Assistance Act."